



Technical Service Bulletin

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Fuel Pump Rebuild

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This Technical Service Bulletin provides procedures to rebuild fuel pumps for the QSK45 and QSK60 Series engine. A new style fuel of pump has been introduced that has a modified shaft diameter to ease rebuild of the pump. Other than this change, the rebuild procedure for old- and new-style pumps is the same. Tables containing lists of common parts used to rebuild the fuel pumps are shown below.

Note : This is a product improvement and is not subject to campaign.

Table 1, Pump Assembly Part Numbers		
Engine Application	Old-Style Pump Assembly Part Number	New-Style Pump and Regulator Assembly Part Number
QSK45 industrial	4001690, 4001701, 4009884 4009885	4062058
QSK60 industrial, QSK60 generator (midflow)	4001694, 4009883	4062059
QSK60 generator (high flow)	4001697, 4009887	4062198

Note : Old- and new-style pumps can be identified by the pump model number. An old-style pump model number ends in FGX, and a new-style pump model number ends in FG.

Table 2, Common Parts Kit			
Kit	Contents	Quantity	Pump Model
Shaft Parts Kit, Part Number 4076763	Shaft	1	40020FG, 40025FG, 40029FG, and 40032FG

Accessories Parts Kit, Part Number 4076764	Hex head capscrews	4	40020FG, 40025FG, 40029FG, 40032FG, 40020FGX, 40025FGX, 40029FGX, and 40032FGX
	Dowel pins	2	
	Check valve	1	
	O-ring	1	
	Pipe plug	1	
Front Cover Parts Kit, Part Number 4087971	Front cover	1	40020FG, 40025FG, 40029FG, and 40032FG
	Bearing sleeve (factory installed into cover)	1	
	Lip seal	1	
	Mechanical seal	1	
	Spacer	1	
	Retaining ring	2	
	Ball bearing	1	
	O-ring	1	
	O-ring	1	
Seal Kit, Part Number 4076478	Lip seal	1	40020FG, 40025FG, 40029FG, 40032FG, 40020FGX, 40025FGX, 40029FGX, and 40032FGX
	Mechanical seal	1	
	Spacer	1	
	Retaining ring	2	
	Ball bearing	1	
	O-ring	1	
	O-ring	1	
Housing Parts Kit, Part Number 4087914	Gear housing	1	40020FG
	Pinion gear	1	
	Ring gear	1	
	Bearing sleeve	2	
	O-ring	1	
	Number 2 drive screw (factory installed into housing)	2	
	Nameplate, serial number (factory installed onto housing)	1	

Housing Parts Kit, Part Number 4076762	Gear housing	1	40025FG
	Pinion gear	1	
	Ring gear	1	
	Bearing sleeve	2	
	O-ring	1	
	Number 2 drive screw (factory installed into housing)	2	
	Nameplate, serial number (factory installed onto housing)	1	
Housing Parts Kit, Part Number 4076761	Gear housing	1	40029FG
	Pinion gear	1	
	Ring gear	1	
	Bearing sleeve	2	
	O-ring	1	
	Number 2 drive screw (factory installed into housing)	2	
	Nameplate, serial number (factory installed onto housing)	1	
Housing Parts Kit, Part Number 4076760	Gear housing	1	40032FG
	Pinion gear	1	
	Ring gear	1	
	Bearing sleeve	2	
	O-ring	1	
	Number 2 drive screw (factory installed into housing)	2	
	Nameplate, serial number (factory installed onto housing)	1	

Table 3, Common Assembly Parts			
Component	Part Number	Quantity	Description
Regulator gasket	3348583	1	For all fuel pump assemblies

Table 4, Common Regulator Parts		
Component	Part Number	Quantity
Actuator	3330600	1
Filter screen	3090769	1
Injector gasket	173086	1
Pressure relief valve	3867737	1

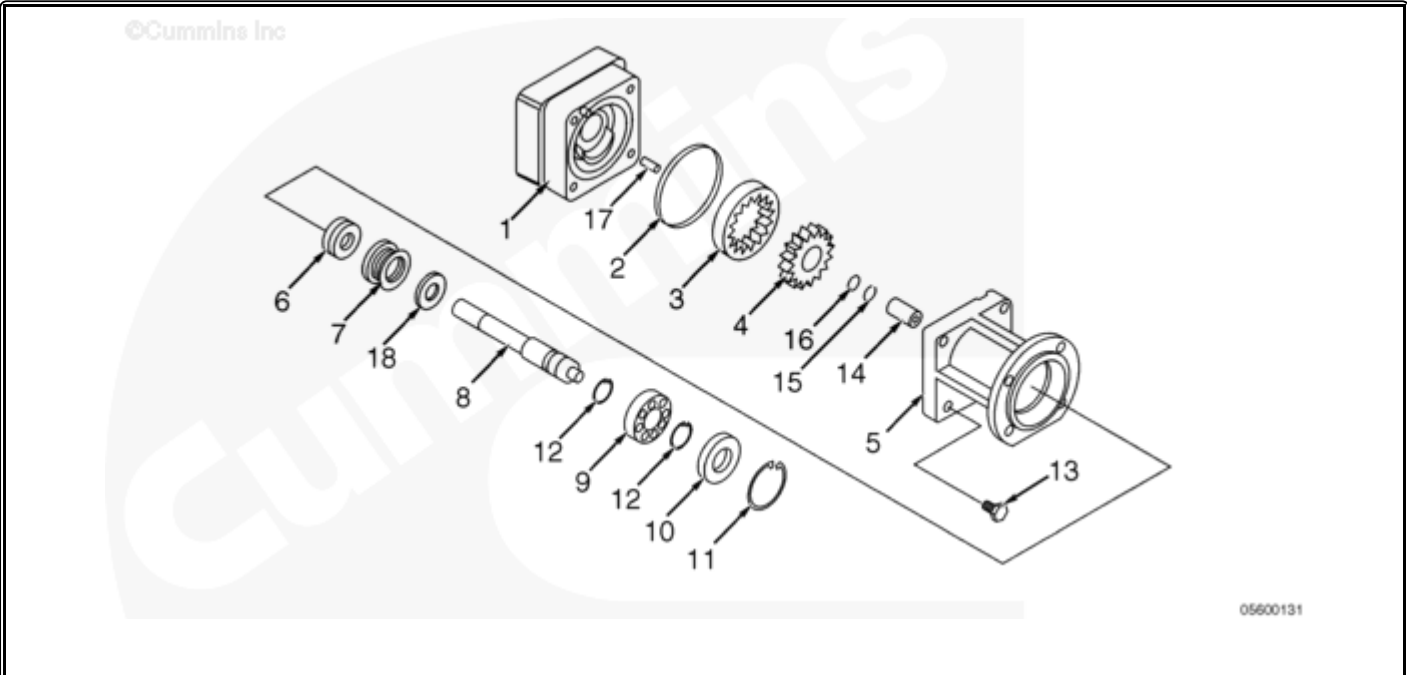


Table 5, Fuel Pump Components		
Item Number	Description	Quantity
1	Gear housing	1
2	Gear housing o-ring	1
3	Ring gear	1
4	Pinion gear	1
5	Front cover	1
6	Mechanical seal stationary ring	1
7	Mechanical seal bellows assembly	1
8	Shaft	1
9	Ball bearing	1

10	Lip seal	1
11	Lip seal retaining ring	1
12	Ball bearing retaining ring	2
13	Capscrew	4
14	Check valve	1
15	Check valve o-ring	1
16	Check valve retaining ring	1
17	Dowel pins	2
18	Spacer	1

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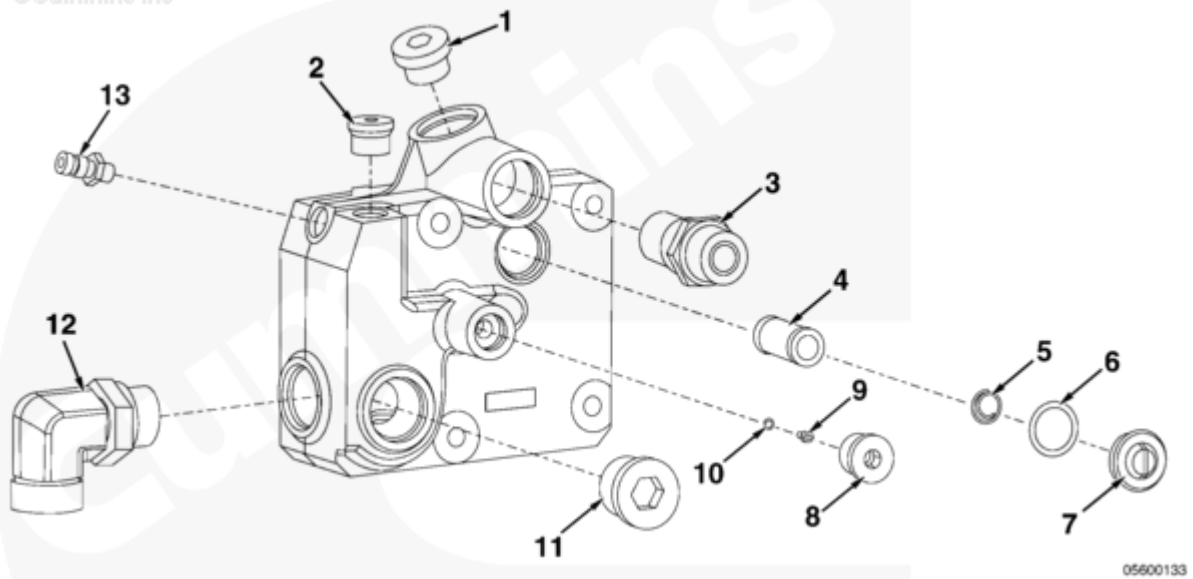


Table 6, Regulator Components (front top view)

Item Number	Description	Quantity
1	Threaded plug	1
2	Threaded plug	1
3	Check valve	1
4	Filter screen	1
5	Compression spring	1
6	O-ring seal	1
7	Fuel pump filter cap	1
8	Threaded plug	1
9	Injector plug orifice	1
10	Injector gasket	1
11	Threaded plug	1

12	Male adapter elbow	1
13	Coupling nipple	1

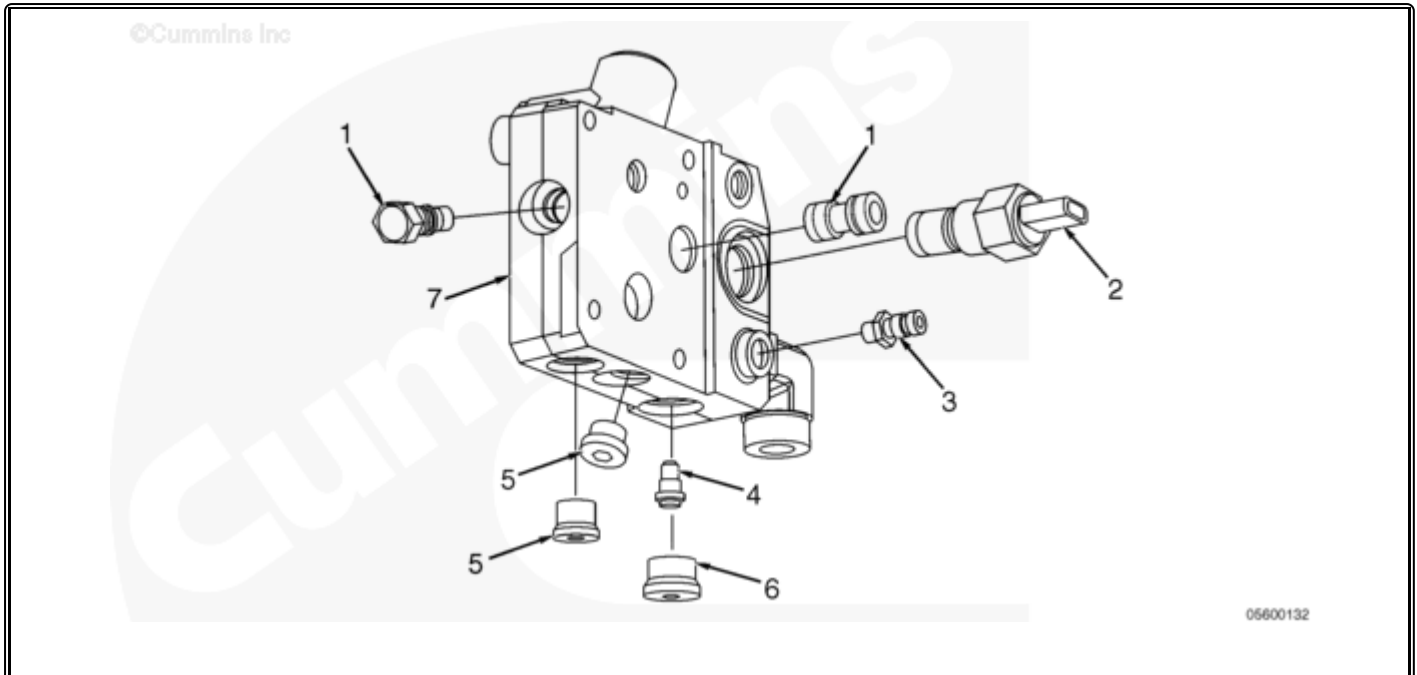


Table 7, Regulator Components (rear bottom view)		
Item Number	Description	Quantity
1	Pressure regulator valve	1
2	Fuel control actuator	1
3	Coupling nipple	1
4	Pressure relief valve	1
5	Threaded plug	2
6	Threaded plug	1
7	Pressure regulator valve body	1